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Simulate tumor microenvironment evolution from sequencing data through a hybrid Multi-Agent Spatio-Temporal model

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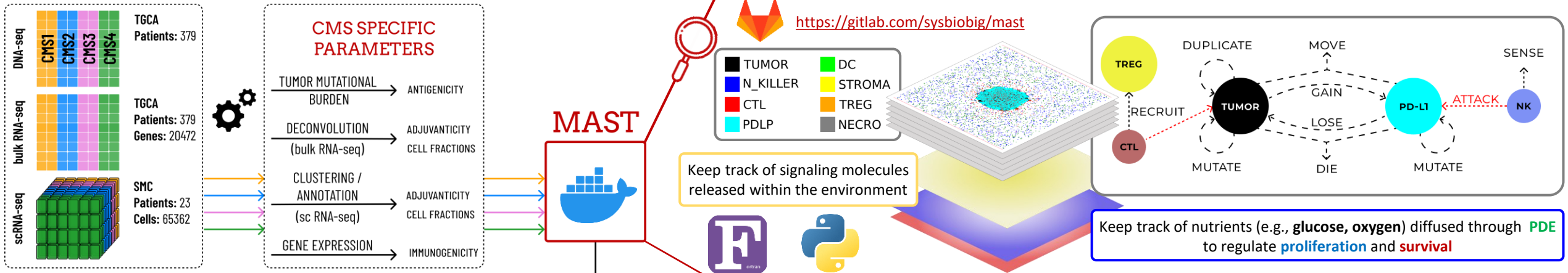
Background

- Different tumor microenvironments (TMEs) composition evolve **differently** in **time** and **space**.
- **Hybrid computational modeling (agent-based model + PDE)** allows a better understanding of **tumor-immune dynamics.**

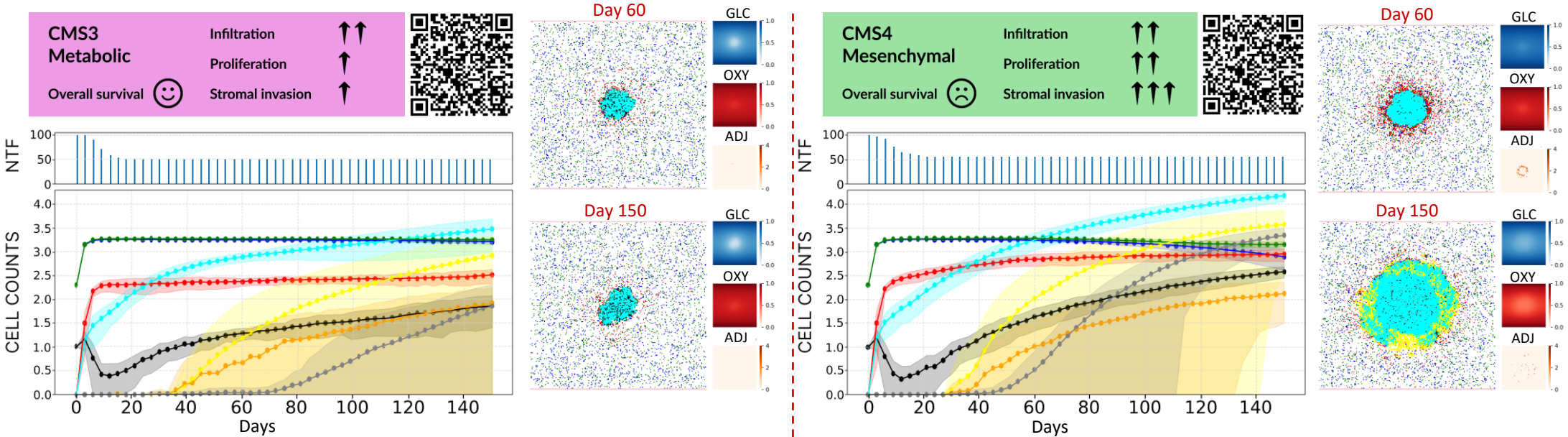
Our work

- Our framework models (1) **spatio-temporal cell dynamics**, (2) **innate and adaptive immune system** responses and (3) **immune escape mechanisms** to mimic TME scenarios, **learning** TMEs parameters from **omics data**.
- We applied our framework to **colorectal cancer scenarios** and its **four consensus molecular subtypes (CMSs)**.

Omics data to inform MAST : a Multi-Agent Spatio Temporal Framework



Colorectal Cancer Simulation Results



Conclusion

- Validation with known **emergent properties**
- **Outcome agreement** with literature
- Implementation of targeted **treatment**

What's next?

- **Scenario-to-patient** specific simulations
- Advanced **treatment investigation**
- Integration of **other omics data**